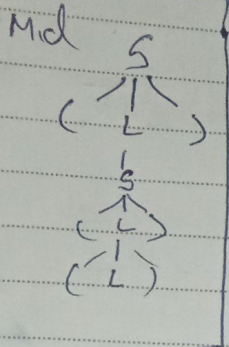


Date: _____

FINAL TERM



<stnt>

18/12/24

Q (if mein else, apne if mein match krna chahiye)

<if> → if exp then stnt | if exp then stnt else stnt | other

if E1 then if E2 then S1 else S2

<stnt>

<if>

(lexeme mein tor kr krna hai)

if

exp

then

stnt

if exp then else

Dangling if

X — X — X — X — X — X — X — X

<stnt>

if

exp

then

stnt

else

stnt

E1

if

exp

then

stnt

S2

E2

S1

(else should match closest if)

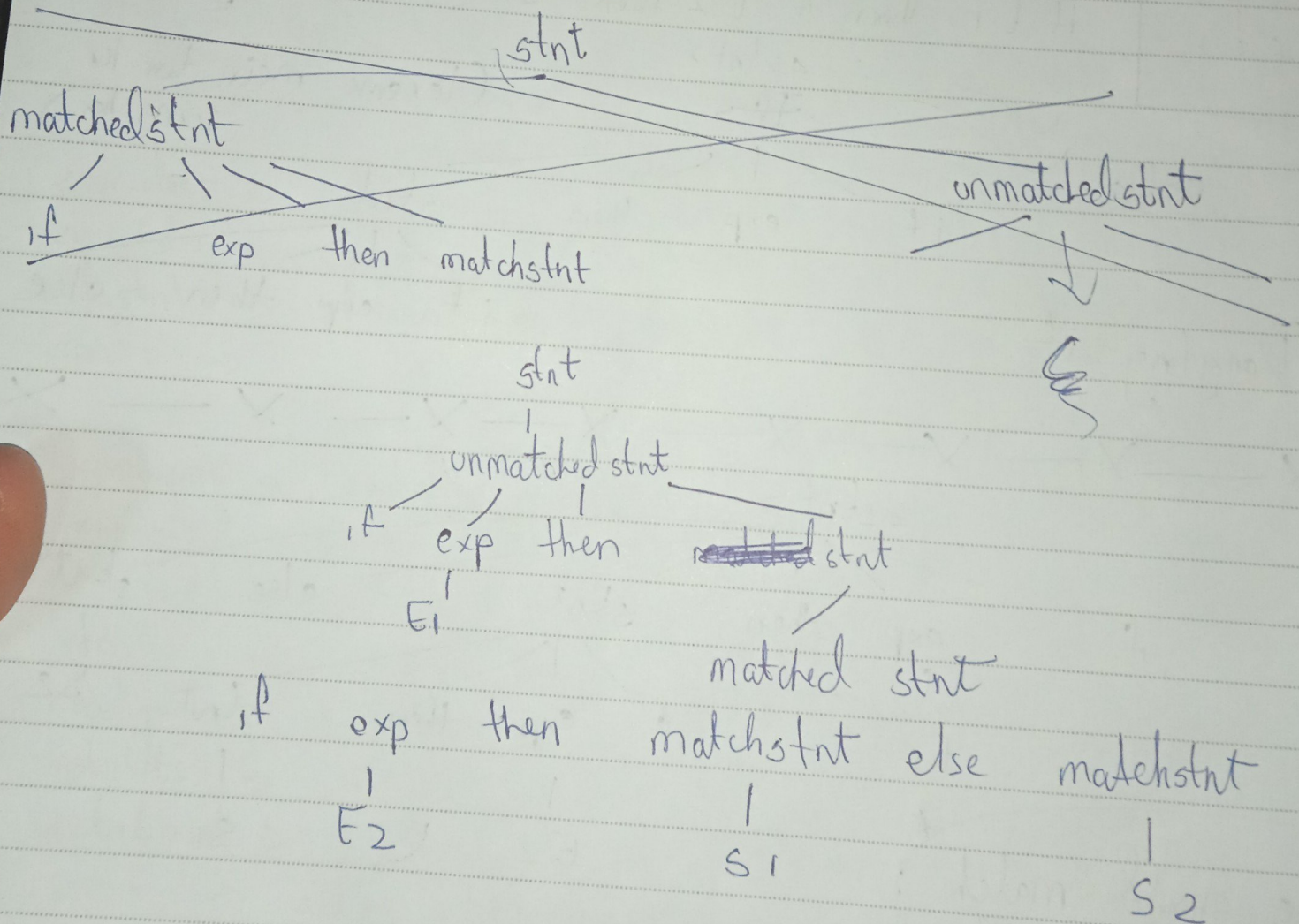
stmt $\rightarrow$ matched stmt | unmatched stmt

Date: _____

match stmt $\rightarrow$ if exp then match stmt else matched stmt | other

unmatched stmt $\rightarrow$ if exp then stmt | if exp then matched stmt else unmatched stmt

if E_1 then E_2 then S_1 else S_2



20/12/24

Q-) Write CFG for a programming language having basic construct like

- ① Multiple Declaration
- ② Conditional Statement
- ③ Loop Construct
- ④ Functions / Pre-defined / User Defined

<program> → begin <stmt> end

<stmt> →

Date: 3/1/25

Top Down, Bottom Up

S

↓

String

↓

Starting Non Terminal

Call

$A \rightarrow AabAab$

↳ Immediate Left Recursion

(No ^T Terminal immediately calling itself)

To remove left recursion

$A \rightarrow A\alpha \mid B$

$A \rightarrow BA'$

$A' \rightarrow \alpha A' \mid \epsilon$

$\{ \text{stmt} \rightarrow \langle \text{opr} \rangle \langle \text{stmt} \rangle \langle \text{ifelse} \rangle \langle \text{stmt} \rangle$
Variability of Length $\rightarrow$ Recursive

$B \rightarrow$ Non Recursive Rule

$\alpha \rightarrow$ Non Recursive part of Recursive Rule

$E \rightarrow E+T \mid T$

$T \rightarrow T*F \mid F$

$F \rightarrow (E) \mid id$

$E \rightarrow \underbrace{E}_A + \underbrace{T}_\alpha \mid \underbrace{T}_B$

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \epsilon \leftarrow$

$T \rightarrow FT'$

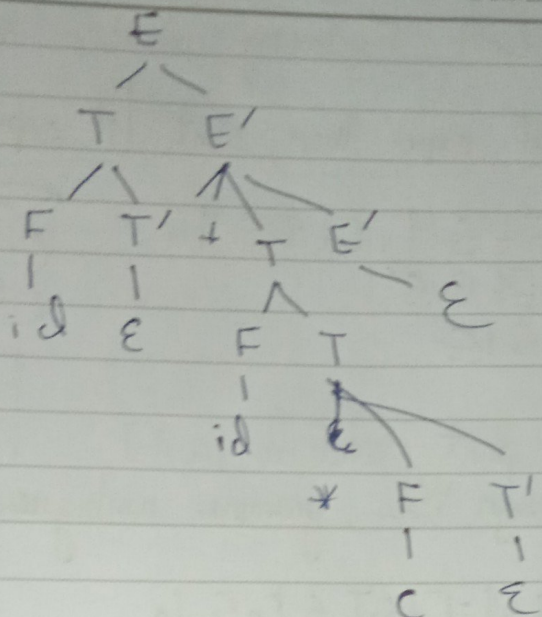
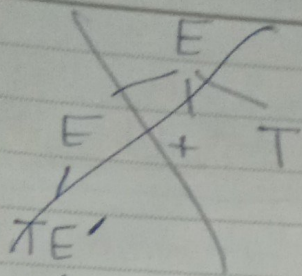
$T' \rightarrow *FT' \mid \epsilon \leftarrow$

~~$F \rightarrow *FT' \mid \epsilon \leftarrow$~~

$F \rightarrow (E) \mid id$

id+id*id

Date: _____



$A \rightarrow A \alpha b c d e$
 α

$A \rightarrow A \alpha_1 b | A \alpha_2 c | A \alpha_3 d | B$ (Scenario 3)

Scenario 1

$A \rightarrow B A'$

$A' \rightarrow \alpha_1 A' | \alpha_2 A' | \alpha_3 A' | \epsilon$

Scenario 2

$A \rightarrow A \alpha | B_1 | B_2 | B_3$

$A \rightarrow B_1 A' | B_2 A' | B_3 A'$

$A' \rightarrow \alpha A' | \epsilon$

$B \rightarrow$ Non Recursive

$B A'$ k saath

Remove Multiple Alpha's & Multiple Beta's

$S \rightarrow (L) | a$

$L \rightarrow L, S | S$

$L \rightarrow S L'$

$L \rightarrow , S L' | \epsilon$

Transition Day, VSL,
Cf + IR + LF + All complete

Date: _____

Left Factoring

$stmt \rightarrow \text{if expr then stmt, | if expr then stmt, else stmt,}$

$stmt \rightarrow E \& S \mid E \& S \& S \mid a$

$stmt \rightarrow E \& S \& S'$

$S' \rightarrow e \& S \mid E$

Common Part remain same

uncommon derive from Non terminal

dec, print, while do, switch
Loop con, design VSL, grammar ambiguous

$stmt \rightarrow \text{if } E \& S \mid \text{if } E \& S \& S \& S \& S \mid a$

(else if)

$stmt \rightarrow E \& S \& S'$

$S' \rightarrow e \& S \& S \& S$

Complete all Construct.

$\langle start \rangle \rightarrow \{ \langle code \rangle \}$
begin $\langle stmt \rangle$ end

$\langle stmt \rangle \rightarrow \langle dec \rangle \mid \langle if else \rangle \& \langle loop \rangle$

$\langle if else \rangle$

$\langle dec \rangle \rightarrow \langle stmt \rangle$

$\langle cond \rangle$

$\langle print \rangle \rightarrow \epsilon$

$\langle start \rangle$
begin $\langle stmt \rangle$ end

$\langle dec \rangle \& \langle stmt \rangle$

Connectivity Imp.

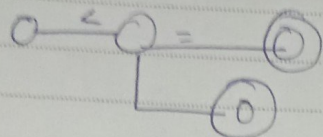
left rec $\rightarrow$ left fact

Recursive Descent Parser

Date: 8/1/24

$S \rightarrow cAd$
 $A \rightarrow ab|a$

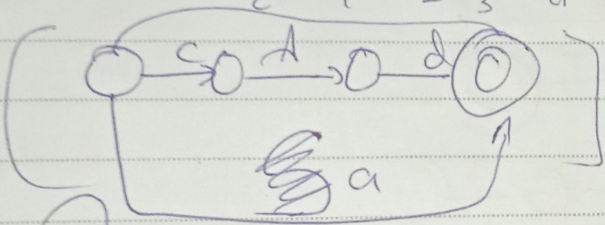
• Lexical Transition Diagram
 ↳ To assign tokens



$\leq \rightarrow 1$ token
 ↳ 1 lexeme

↳ Designing them for Grammar

$S \rightarrow cAd|a|\epsilon|$
 $\epsilon \quad 1 \quad 2 \quad 3 \quad 4$



Representing S

Parsing → ① Top Down
 ↳ Bottom Up

Top Down

S

↓

a

Bottom Up

abcd

↓

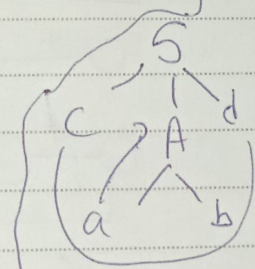
S

Back Tracking Structure

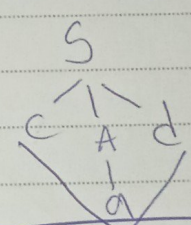
$S \rightarrow cAd$
 $A \rightarrow ab|a$

$S \rightarrow cAd$
 $A \rightarrow ab|a$

cad



cabcd
 we need
 cad



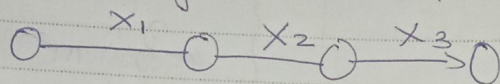
Date: _____

Code $\rightarrow$ Lexeme $\rightarrow$ Terminal

How to make Transition Diagram?
 $\times$ Eliminate Left Recursive + Left factor + Initial State
 + Final States $\rightarrow$ Connect Path from initial to final.
 (Book Notation)

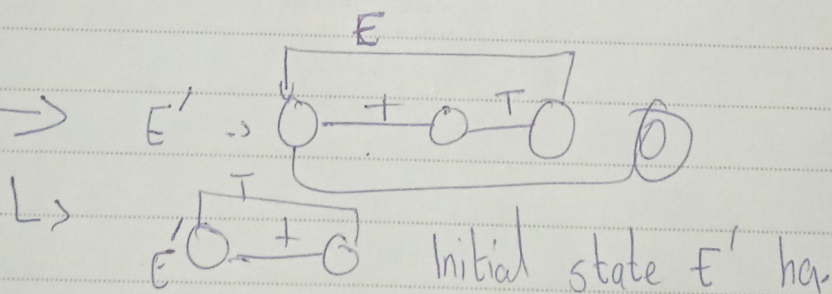
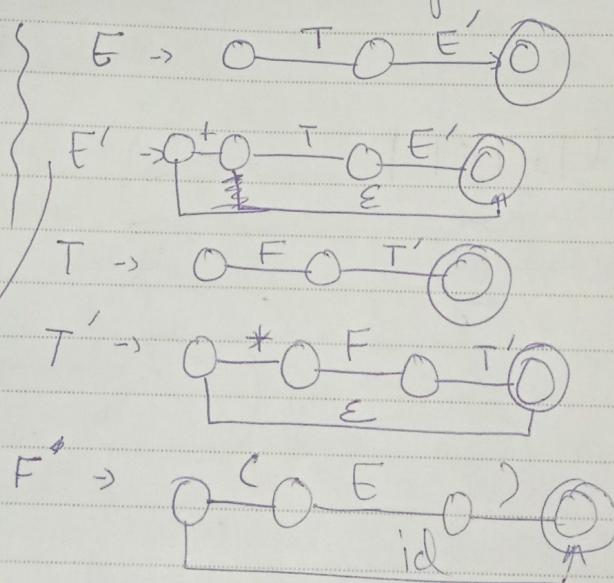
$A \rightarrow X_1 X_2 X_n$

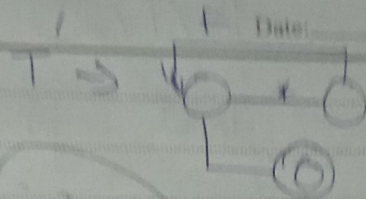
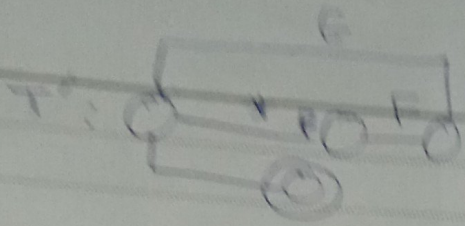
Non Terminal k lie Initial & Final State



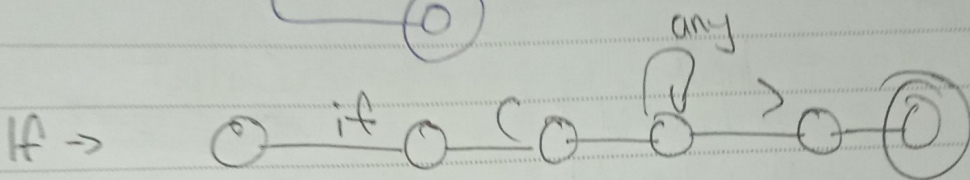
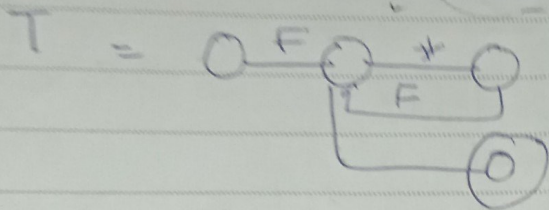
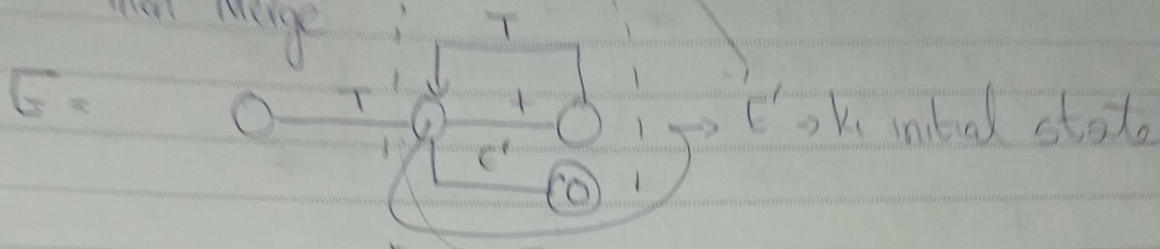
Transition $\rightarrow$ Some Non Terminal k lie Transition Diagram
 Separate Diagram

$E \rightarrow TE'$
 $E' \rightarrow +TE'/\epsilon$
 $T \rightarrow FT'$
 $T' \rightarrow *FT'/\epsilon$
 $F' \rightarrow (E)id$





Then Merge



10/1/25

String $\rightarrow$ Table

Date: _____

Input Symbol

Non-Terminal	id	+	*	(	)	\$
E	$E \rightarrow TE'$	$E + TE'$		$E \rightarrow TE'$	$E' \rightarrow E$	$E' \rightarrow E$
T	$T \rightarrow FT'$			$T \rightarrow FT'$		
T'		$T' \rightarrow E$	$T' \rightarrow *FT'$		$T' \rightarrow E$	$T' \rightarrow E$
F	$F \rightarrow id$		$F \rightarrow (E)$			

Predictive Parsing Table

$\$ \rightarrow end$

Stack

$\$E$
 $\$E'T$
 $\$E'T'F$
 $\$E'T'id$
 $\$E'T'$
 $\$E'$
 $\$E'T$

Input

$id + id * id \$$
 $id + id * id \$$
 $id + id * id \$$
 $id + id * id \$$
 $+ id * id \$$
 $+ id * id \$$
 $id * id \$$

Output

~~$E \rightarrow TE'$~~
 $E \rightarrow TE'$
 $T \rightarrow FT'$
 $F \rightarrow id$
 $T' \rightarrow E$
 $E' \rightarrow +TE'$

$\hookrightarrow$ Right most se le kr chalna hq

Date: _____

Non-Terminal	id	+	*	(	)	\$
E	$E \rightarrow TE'$			$E \rightarrow TE'$		
E'		$E' \rightarrow TE'$			$E' \rightarrow E$	$E' \rightarrow E$
T	$T \rightarrow FT'$			$T \rightarrow FT'$		
T'		$T' \rightarrow E$	$T' \rightarrow *FT'$			
F	$F \rightarrow id$			$F \rightarrow (E)$		

Stack

\$E
\$E'T
\$E'T'F
\$E'T'id
\$E'T'
\$E'
\$E'T
\$E'T'F
\$E'T'F
\$E'T'id
\$E'T'
\$E'T'F*
\$E'T'F
\$E'T'id
\$E'T'
\$E'
\$

Input

id+id*id\$
id+id*id\$
id+id*id\$
~~id+id*id\$~~
+id*id\$
+id*id\$
~~id*id\$~~
id*id\$
~~id*id\$~~
*id\$
~~*id\$~~
id\$
~~id\$~~
\$
\$
\$

Output

$E \rightarrow TE'$
 $T \rightarrow FT'$
 $F \rightarrow id$
 $T \rightarrow E$

 $E' \rightarrow TE'$
 $T \rightarrow FT'$
 $F \rightarrow id$

 $T' \rightarrow *FT'$

 $F \rightarrow id$

 $T' \rightarrow E$
 $E' \rightarrow E$

$E \rightarrow TE'$
 $E' \rightarrow +TE' \mid \epsilon$
 $T \rightarrow FT'$
 $T' \rightarrow *FT' \mid \epsilon$
 $F \rightarrow (E) \mid id$

Date: _____

funct for Table

First (A)
 Follow (A)

$First(F) \rightarrow \{ (, id \}$
 $First(E) \rightarrow \{ (, id \}$
 $First(T) \rightarrow \{ \}$

$First(E) = First(T) = First(\epsilon F) = \{ (, id \}$
 $First(E') = \{ +, \epsilon \}$
 $First(T') = \{ *, \epsilon \}$

Follow

$\{ \$, > \}$
 $Follow(E) \Rightarrow \{ >, \$ \}$

Non T ko Non T
 follow

Follow (T)

$\{ +, >, \$ \}$

\$ ko add

$Follow(T) = First(E') + Follow(E)$
 $\{ +, >, \$ \}$

→ First of any Non Terminal is the first Terminal that is ~~not~~ read from the Non Terminal.

→ First of any Non Ter → is Non Term → first NT ka first

15/1/25

$$\text{First}(E') = \{+, \epsilon\}$$

$$\text{First}(T') = \{*, \epsilon\}$$

Follow(T) = Follow(T) coz nothing is following T'

$$\text{Follow}(F) = \{*, +, \epsilon, \$\}$$

$$\text{Follow}(E') = \{\$, \epsilon\}$$

LL(1)

	id	+	*	(	)	\$
E	$E \rightarrow TE'$			$E \rightarrow TE'$		
E'		$E' \rightarrow TE'$			$E \rightarrow \epsilon$	$E' \rightarrow \epsilon$
T	$T \rightarrow FT'$			$T \rightarrow FT'$		
T'		$T' \rightarrow \epsilon$	$T' \rightarrow *FT'$		$T' \rightarrow \epsilon$	$T' \rightarrow \epsilon$
F	$F \rightarrow id$					

Predictive Parsing Table

$\$ \rightarrow$ Starting N-T

Date: _____

First $\rightarrow$

- Follow $\rightarrow$
- ① Add $\$$ of starting Non Ter
 - ② If N-T is following N-T A, so the follow of A that 1 Ter, the first of the following N.

if $\rightarrow$ Follow(A) = First of (Following NT)

Follow(A) First of B

- ③ If first(B) includes NULL ϵ , then follow of its originator is includes too.
- $\rightarrow$ Nothing is following N-T $\rightarrow$ Originator

Q2)

$$S \rightarrow iEtSS' | a$$

$$S' \rightarrow eS | \epsilon$$

$$E \rightarrow b$$

$$\begin{aligned}\text{First}(S) &\rightarrow \{i, a\} \\ \text{First}(S') &\rightarrow \{e, \epsilon\} \\ \text{First}(E) &\rightarrow \{b\}\end{aligned}$$

$$\begin{aligned}\text{Follow}(S) &\rightarrow \{\$, e\} \\ \text{Follow}(S') &\rightarrow \{e, \$\} \\ \text{Follow}(E) &\rightarrow \{t\}\end{aligned}$$

Design P. Parsing

$\times LL(1) \times \downarrow$

Date: _____

	i	t	a	be	\$ b	\$
S	$S \rightarrow iEtSS'$		$S \rightarrow a$	$S \rightarrow e$		
S'				$S' \rightarrow eS$		$S' \rightarrow \&\$$
E					$E \rightarrow b$	

$\rightarrow$ 1 cell mein

1 NT ko doosra NT follow kr rha hai

2 $\rightarrow$ Ambiguous

$E \rightarrow$ follow of Originator

Row $\rightarrow$ Non Terminal

Column $\rightarrow$

LL(1) Grammar

$\rightarrow$ left 2 right scanning
 $\rightarrow$ left most derivation

(LL(K)) $\rightarrow$ Look ahead

LL(1) $\rightarrow$ Any Multi Define

BOTTOM UP PARSING

Date: _____

$S \rightarrow aABe$
 $A \rightarrow Abc \mid b$
 $B \rightarrow d$

Handle

$A \rightarrow b$
 $B \rightarrow d$

$abbcde$
 $aAbcde$

$aAde$
 $aABe$
 S

String

Starting Non Terminal

Called Bottom Up Approach

Right Most

$S \rightarrow aABe$
 $aAde$
 $aAbcde$
 $abbcde$

reduce

Handles $\rightarrow$ Substring Matching $\rightarrow$ Right side of Rule

Ambiguous Nahi honi Chahiye Top down / Bottom Up

Bottom

→ Shift Reduce Parser
↳ LR Parser

↳ LR(0) → Lookup
SLR Parser

Date: 17/1/25

• k bad
NT → uske rule
seare from I_0

$E' \rightarrow \cdot E$

$E \rightarrow \cdot E + T$

$E \rightarrow \cdot T$

$T \rightarrow \cdot T * F$

$T \rightarrow \cdot F$

$F \rightarrow \cdot (E)$

$F \rightarrow \cdot id$

Goto (I_0, E)

$\left[\begin{array}{l} E' \rightarrow E. \\ E \rightarrow E. + T \end{array} \right] I_1$

goto (I_0, T)

$\left[\begin{array}{l} E \rightarrow T. \\ T \rightarrow T. * F \end{array} \right] I_2$

Goto (I_0, F)

$T \rightarrow F.] I_3$

Go to ($I_0, ($)

$F \rightarrow (\cdot E$

$E \rightarrow \cdot E + T$

$E \rightarrow \cdot T$

$T \rightarrow \cdot T * F$

$T \rightarrow \cdot F$

$F \rightarrow \cdot (E)$

$F \rightarrow \cdot id$

I_4

Canonical

Date: _____

(I_0, id)

$(F \rightarrow id \cdot)] I_5$

$(I_1, +)$

$E \rightarrow E + \cdot T$

~~$T \rightarrow \cdot T * F$~~

$T \rightarrow \cdot F$

$F \rightarrow \cdot (E)$

$F \rightarrow \cdot id$

I_6

$(I_2, *)$

$T \rightarrow T * \cdot F$

$F \rightarrow \cdot (E)$

$F \rightarrow \cdot id$

I_7

(I_4, E)

$F \rightarrow (E \cdot)$

$E \rightarrow E + \cdot T$

I_8

(I_4, T)

$E \rightarrow T \cdot$

$T \rightarrow T \cdot * F$

I_2

(I_4, f)

$(T \rightarrow f \cdot)$

I_3

$$(I_4, ())$$

$$- F \rightarrow (.E)$$

$$E \rightarrow$$

$$E \rightarrow$$

$$T \rightarrow$$

$$T \rightarrow$$

$$F \rightarrow$$

$$F \rightarrow$$

$$(I_4, id)$$

$$F \rightarrow id.$$

$$I_5$$

$$I_9 \rightarrow$$

$$(I_6, T)$$

$$\begin{bmatrix} E \rightarrow E + T. \\ T \rightarrow T * F \end{bmatrix}$$

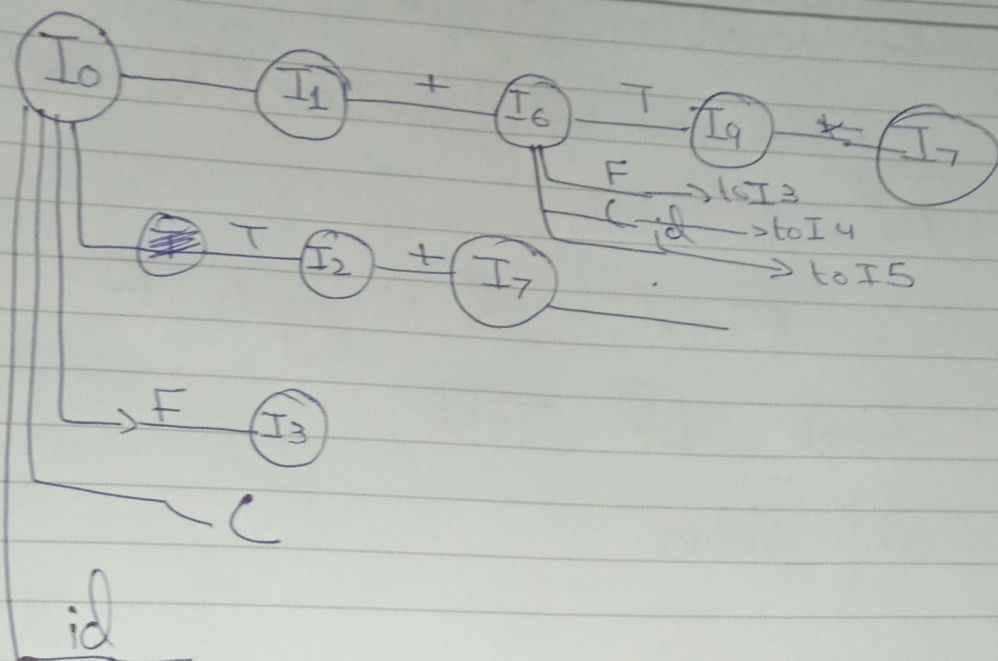
$$I_9$$

$$(I_6, F)$$

$$[T \rightarrow F.]$$

$$I_3$$

$$(I_6, ())$$



$E \rightarrow E + T$
 $E \rightarrow T$
 $T \rightarrow T * F$
 $T \rightarrow F$
 $F \rightarrow (E)$
 $F \rightarrow id$

First Follow (E) $\rightarrow (, id$
 First (T) $\rightarrow (, id$
 First (F) $\rightarrow (, id$
 Follow (E) $\rightarrow (, +, \$$
 Follow (T) $\rightarrow *, +, >, \$$
 Follow (F) $\rightarrow +, >, \$$

F ko koi follow nhi kr raha, Originator ke follow lkh deien

Pen Check

Date: _____

Go To Action Table

• bad mein aye Production Rule complete.

R4 → Reduce to Rule 4

F → id id ki jagah F

Simple LR Table

Shift Reduce